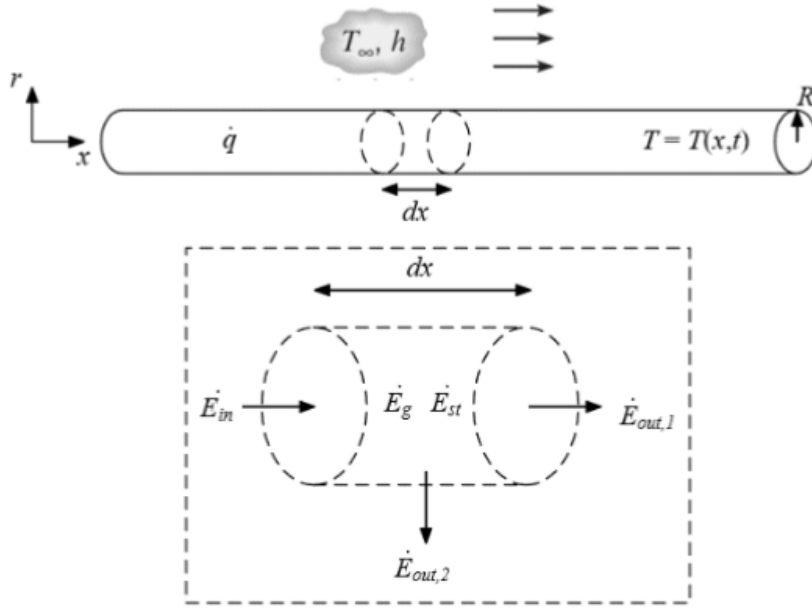


PROBLEM 1

KNOWN: Rod dimensions: R, L ; environment: T_∞, h ; volumetric heat generation rate: $\dot{q}$; $T(x=0) = T_1$ for all t ; $T(t=0) = T_i$ for all x .

FIND: (a) For a differential cylinder, write down energy balance and express each term as a function of T ; (b) Derive governing equation; (c) Express boundary conditions and initial conditions

SCHEMATIC:



ASSUMPTIONS: Very long rod, negligible thermal radiation, uniform radial temperature, i.e., $T = f(x, t)$, constant thermal and mechanical properties k, ρ , and c_p .

ANALYSIS: Part (a): The energy balance for a control volume is:

$$\begin{aligned}\dot{E}_{in} - \dot{E}_{out} + \dot{E}_g &= \dot{E}_{st} \\ \dot{E}_{in} - (\dot{E}_{out,1} + \dot{E}_{out,2}) + \dot{E}_g &= \dot{E}_{st}\end{aligned}$$

Let $\Delta v \equiv A dx$ where $A = \pi R^2$ and dx is the length of the differential control volume. Each term is:

$$\dot{E}_{in} = q_{cond} = \boxed{-kA \frac{dT}{dx}} \quad (1)$$

$$\dot{E}_{out,1} = q_{cond} = -kA \frac{dT}{dx} + \frac{d}{dx} \left(-kA \frac{dT}{dx} \right) dx = \boxed{-kA \left(\frac{dT}{dx} + \frac{d^2T}{dx^2} dx \right)} \quad (2)$$

$$\dot{E}_{out,2} = q_{conv} = hP dx (T_s - T_\infty) = \boxed{hP dx (T - T_\infty)} \quad (3)$$

$$\dot{E}_g = \dot{q} \Delta v = \boxed{\dot{q} A dx} \quad (4)$$

$$\dot{E}_{st} = \rho c_p \frac{dT}{dt} \Delta v = \boxed{\rho c_p \frac{dT}{dt} A dx} \quad (5)$$

Part (b): Substitute the individual terms into the energy balance and simplify:

$$\begin{aligned}
 \dot{E}_{in} - \dot{E}_{out,1} - \dot{E}_{out,2} + \dot{E}_g &= \dot{E}_{st} \\
 -kA \frac{dT}{dx} - \left(-kA \left(\frac{dT}{dx} + \frac{d^2T}{dx^2} dx \right) \right) - (hP dx (T - T_\infty)) + (\dot{q} A dx) &= \rho c_p \frac{dT}{dx} A dx \\
 k \frac{d^2T}{dx^2} \cancel{A dx} - hP dx (T - T_\infty) + \dot{q} \cancel{A dx} &= \rho c_p \frac{dT}{dx} \cancel{A dx} \\
 k \frac{d^2T}{dx^2} - \frac{hP dx}{\Delta v} (T - T_\infty) + \dot{q} &= \rho c_p \frac{dT}{dx} \\
 \frac{hP dx}{\Delta v} &= \frac{h \cdot 2\pi R \cdot dx}{\pi R^2 dx} = \frac{2h}{\pi R} \\
 \boxed{k \frac{d^2T}{dx^2} + \frac{2h}{\pi R} (T - T_\infty) + \dot{q} = \rho c_p \frac{dT}{dx}}
 \end{aligned}$$

Part (c): We have a Dirichlet boundary condition at $x = 0$ and a Neumann boundary condition at $x = L$:

$$\boxed{T(x = 0, t) = T_1} \quad (6)$$

$$\text{at left end: } q''_{conv} = q''_{cond} \implies -k \frac{dT}{dx} = h(T(x = L, t) - T_\infty) \implies \boxed{\left. \frac{dT}{dx} \right|_{x=L} = -\frac{h}{k}(T - T_\infty)} \quad (7)$$

The initial boundary condition can be expressed as $\boxed{T(x, t = 0) = T_i}$